Making deep learning work

Unit 3 · reference sheet · AI/ML in 35 topics

Regularisers

Dropout: zero w.p. p, rescale $\frac{1}{1-p}$; **identity at inference**
Weight decay: $L + \lambda\|w\|^2$ Early stop: keep best val
Augmentation must preserve the label (flip $6 \leftrightarrow 9$: no)

Signal through depth

gain g over L layers $\Rightarrow g^L$
$0.25^{10} = 9.5 \times 10^{-7}$ $1.5^{10} = 58$ $1.5^{50} = 6.4 \times 10^{8}$
Fixes: He/Xavier init, LayerNorm, residual $y = x + f(x)$, clip, warmup

Architecture priors

Family	Assumes	Cost
MLP	no structure	$n_\ell n_{\ell-1}$
CNN	locality + translation	$k^2 C_{in} C_{out}$
RNN	order, shared dynamics	sequential, $O(T)$

Conv $3 \times 3 \times 64 \times 64 = 36{,}864$ vs dense 205M on 224^2.

Gotchas

- BatchNorm at inference uses running stats: call `eval()`.
- Depth without residuals can *reduce* trainability.
- CNN on tabular data: adjacency is arbitrary, prior is false.

This unit at a glance

#	Topic	You can
12	Overfitting in DL	pick among dropout, weight decay, augmentation, early stopping for a given sym
13	Optimization failure modes	diagnose vanishing/exploding gradients; name the fix (init, norm, residual, cl
14	Architecture families	state what MLP / CNN / RNN each assume about data, and choose one

If you see this → do that

Signal	Move
Few images	pretrained CNN
Short sequences	1-D CNN / GRU
Long-range dependency	attention
Tabular data	GBM first
Tight latency	distil / quantise

Numbers worth memorising

$0.25^{50} = 7.9 \times 10^{-31}$ (below float32)
$1.5^{50} = 6.4 \times 10^{8}$
Conv 3×3, 64 in/out: 36,864 vs dense 205M
Dropout $p = 0.4 \Rightarrow$ rescale by $1/0.6 = 1.67$